

# AGAN AGAN

## AI / ML Engineer | Full-Stack Developer | Android Developer

Email: [aagan9904@gmail.com](mailto:aagan9904@gmail.com)

Phone: +254793291158

Location: Bungoma, Kenya

GitHub: [github.com/aagan9904-a11y](https://github.com/aagan9904-a11y)

LinkedIn: [linkedin.com/in/agan-agan-8862b2266](https://linkedin.com/in/agan-agan-8862b2266)

## Professional Summary

Aspiring Software Engineer and Machine Learning practitioner with strong hands-on experience in building deep learning models using PyTorch and deploying real-world AI applications using Flask. Experienced in full-stack development with Django and React, Android development using Kotlin, and Linux-based development environments. Passionate about building scalable AI systems, solving real-world problems, and continuously improving software engineering skills through practical project development and research.

## Technical Skills

**Programming Languages:** Python, Kotlin, Bash, JavaScript, HTML, CSS

**Machine Learning & AI:** PyTorch, CNNs, Neural Networks, Deep Learning, Computer Vision, NLP, Image Classification, Data Preprocessing

**Web Development:** Flask, Django, React, REST APIs, Frontend & Backend Development

**Mobile Development:** Kotlin, Android Studio, Firebase

**Tools & Platforms:** Git, GitHub, Linux (Kali Linux), VS Code, Android Studio

**Databases:** SQLite, Firebase Database

## Experience

### Machine Learning Engineer (Self-Directed Projects)

- Built and trained CNN models for handwritten digit recognition.
- Improved model performance through training optimization and experimentation.
- Developed Flask applications for AI model deployment and real-time prediction systems.

### Full-Stack Developer (Learning & Projects)

- Built backend systems using Django framework.
- Developed interactive frontend interfaces and integrated APIs.
- Structured projects using industry-standard development practices.

### Android Developer

- Developed mobile applications using Kotlin and Android Studio.
- Integrated Firebase backend services and external APIs.

### Cybersecurity & Linux Practice

- Worked extensively in Kali Linux environments.
- Practiced networking fundamentals, Linux administration, and security concepts.

## ■ Projects

### ■ *AI-Powered Handwritten Digit Image Classifier*

Designed and implemented a CNN-based image classification system using PyTorch and the MNIST dataset. Built preprocessing pipelines, optimized model accuracy, and deployed the trained model through a Flask web application for real-time predictions. Demonstrated expertise in deep learning, computer vision, model evaluation, and AI deployment.

### ■ *AI Spam Detection System*

Developing a machine learning system capable of classifying incoming messages as spam or legitimate. Applied NLP techniques such as text cleaning, tokenization, and feature extraction while building the backend architecture with Django. Designed model training and prediction workflows for accurate message classification.

### ■ *Android Translation Application*

Built a mobile translation application using Kotlin and Android Studio. Integrated translation APIs and Firebase services while implementing voice-input functionality. Focused on responsive design and user-friendly mobile experiences.

### ■ *Personal Developer Portfolio Website*

Created a responsive portfolio website showcasing technical projects, achievements, skills, downloadable CVs, and professional contact information. Implemented using HTML, CSS, JavaScript, and GitHub Pages deployment.

### ■ *Machine Learning Learning Lab*

Built a collection of machine learning experiments covering regression, classification, neural networks, CNNs, model evaluation, and data preprocessing. Used as a practical environment for learning and testing AI concepts.

## Education

### Kibabii University

Bachelor of Information Technology (BIT)

Final Year Student (Expected Graduation: 2027)

## Interests

Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Full-Stack Development, Mobile Application Development, Open Source Technologies, Cybersecurity, Cloud Computing, and Software Engineering.

## Future Project Roadmap

- Self-Driving Car Simulation using Computer Vision, OpenCV, Deep Learning, and Reinforcement Learning.
- Face Recognition System using OpenCV and CNN-based real-time detection.
- AI-Powered Hospital Management System using Django, React, and Machine Learning Analytics.
- Student Performance Predictor using Neural Networks, Flask, and Data Visualization.